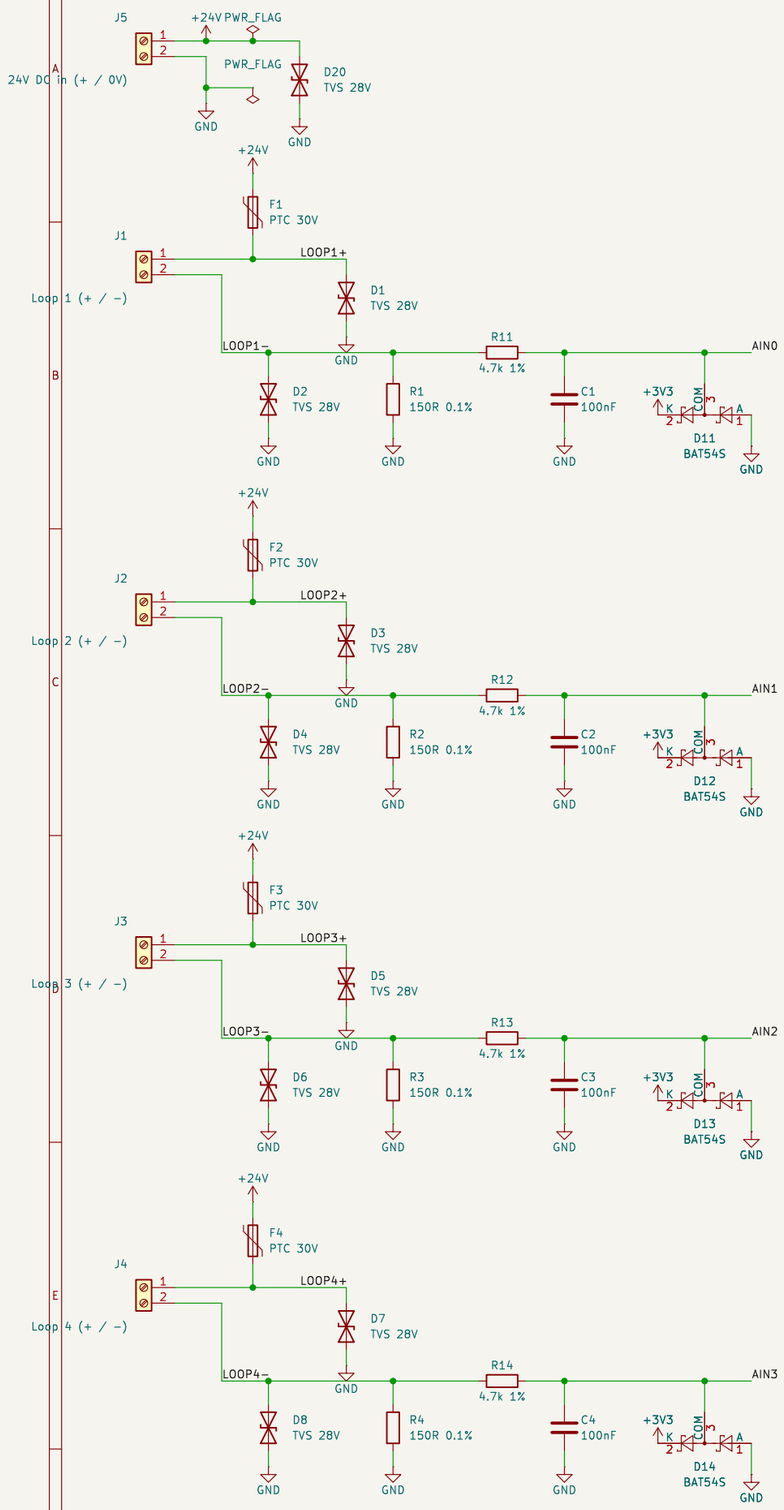
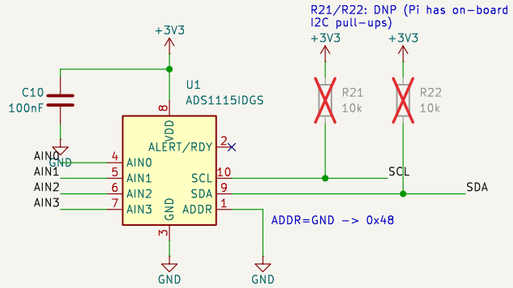
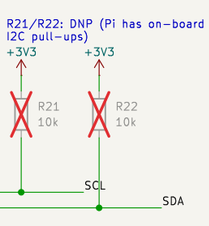
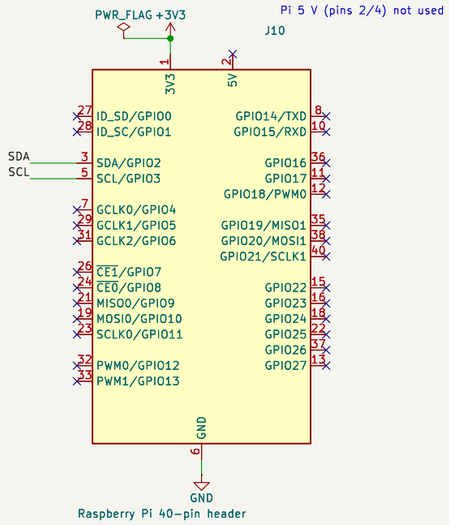








4-channel 4-20 mA input board for the Raspberry Pi 40-pin header - UNTESTED REFERENCE DESIGN
Each channel: PTC -> field terminal (+) -> transmitter -> field terminal (-) -> 150 R shunt -> 0 V,
with TVS on both field wires, 4.7k/100n RC filter and BAT54S clamp in front of the ADS1115 input.
ADS1115 at 0x48 (ADDR = GND); move ADDR to VDD/SDA/SCL for 0x49/0x4A/0x4B to stack boards.
One external 24 V supply feeds all loops through J5; its 0 V is the Pi GND (non-isolated design).
Not included: HAT+ ID EEPROM, mounting/keep-out, reverse-polarity protection, isolation,
GPIO-switchable 3.3 V rail for U1 (recommended for I2C recovery after a brown-out).

mqttpi-4-20ma - MIT licensed
Monitoring/telemetry only - not a certified gas-detection controller
UNTESTED reference design - no board has been built from this file

Sheet: /
File: ads1115-4-20ma-hat-4ch.kicad_sch

Title: 4-channel 4-20 mA Pi HAT - reference design (untested)

Size: A3	Date: 2026-10-27	**Rev: 0.1**
KiCad E.D.A. 9.0.2		Id: 1/1